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Roll No: 2403A52L06

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech	**Assignment Type: Lab**	**Academic Year:**2025-2026	
Course Coordinator Name	Dr. Rishabh Mittal		
Instructor(s) Name	Mr. S Naresh Kumar Ms. B. Swathi Dr. Sasanko Shekhar Gantayat Mr. Md Sallauddin Dr. Mathivanan Mr. Y Srikanth Ms. N Shilpa Dr. Rishabh Mittal (Coordinator) Dr. R. Prashant Kumar Mr. Ankushavali MD Mr. B Viswanath Ms. Sujitha Reddy Ms. A. Anitha Ms. M.Madhuri Ms. Katherashala Swetha Ms. Velpula sumalatha Mr. Bingi Raju		
Course Code	23CS002PC304	**Course Title**	AI Assisted Coding
Year/Sem	III/I	**Regulation**	R23
Date and Day of Assignment	Week 2 - Wednesday	**Time(s)**	23CSBTB01 To 23CSBTB52
Duration	2 Hours	**Applicableto Batches**	All batches
Assignment Number: 3.3(Present assignment number)/**24**(Total number of assignments)			

Q.No.	Question	*Expected Time to complete*
1	**Lab 4: Advanced Prompt Engineering – Zero-shot, One-shot, and Few-shot Techniques** **Lab Objectives** • To explore and apply different levels of prompt examples in AI-assisted code generation • To understand how zero-shot, one-shot, and few-shot prompting affect AI output quality • To evaluate the impact of context richness and example quantity on AI performance • To build awareness of prompt strategy effectiveness for different problem types	Week2 - Wednesday

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Lab Outcomes (LOs)
After completing this lab, students will be able to:

- Use zero-shot prompting to instruct AI with minimal context
- Use one-shot prompting with a single example to guide AI code generation
- Apply few-shot prompting using multiple examples to improve AI responses
- Compare AI outputs across different prompting strategies

Task 1: Zero-Shot Prompting – Leap Year Check
Scenario
Zero-shot prompting involves giving instructions without providing examples.
Task Description
Use zero-shot prompting to instruct an AI tool to generate a Python function that:

- Accepts a year as input
- Checks whether the given year is a leap year
- Returns an appropriate result

Note: No input-output examples should be provided in the prompt.
Expected Output

- AI-generated leap year checking function
- Correct logical conditions
- Sample input and output
- Screenshot of AI-generated response (if required)

Prompt:
Write a Python program where you need to enter an input as year and should check whether it is a leap year or non leap year.
Code:

lab assignment4.3.py > ...

```python
def is_leap_year(year):
    if (year % 400 == 0) or (year % 4 == 0 and year % 100 != 0):
        return "Leap Year"
    else:
        return "Not a Leap Year"


year = int(input("Enter a year: "))
result = is_leap_year(year)
print(result)
```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL POR

```
PS C:\Users\sanjay\AI Assistant coding> & C:\Us
oding/lab assignment4.3.py"
Enter a year: 2020
Leap Year
PS C:\Users\sanjay\AI Assistant coding> & C:\Us
oding/lab assignment4.3.py"
Enter a year: 2025
Not a Leap Year
PS C:\Users\sanjay\AI Assistant coding>
```

Approach:-
In this program we can learn that how zero short prompting works in leap year program without giving an example.

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Task 2: One-Shot Prompting – Centimeters to Inches Conversion
Scenario
One-shot prompting guides AI using a single example.
Task Description
Use one-shot prompting by providing one input-output example to generate a Python function that:

- Converts centimeters to inches
- Uses the correct mathematical formula

Example provided in prompt:
Input: 10 cm → Output: 3.94 inches
Expected Output

- Python function with correct conversion logic
- Accurate calculation
- Sample test cases and outputs

Prompt:
Write a Python program that converts centimeters to inches. Example:input should be 10 cm and output should be 3.94 inches.
Code:

```python
#task2
cm = float(input("Enter length in centimeters: "))
inches = cm / 2.54
print(f"{inches:.2f} inches")
```

Output:

```
PS C:\Users\sanjay\AI Assistant coding> & C:\Users
oding/lab assignment4.3.py"
Enter length in centimeters: 12.3
4.84 inches
PS C:\Users\sanjay\AI Assistant coding>
```

Approach:
In this python program we can learn that how one Short promting involves by giving an example in the prompt**.**

Task 3: Few-Shot Prompting – Name Formatting
Scenario
Few-shot prompting improves accuracy by providing multiple examples.
Task Description
Use few-shot prompting with 2–3 examples to generate a Python function that:

- Accepts a full name as input
- Formats it as “Last, First”

Example formats:

- "John Smith" → "Smith, John"
- "Anita Rao" → "Rao, Anita"

Expected Output

- Well-structured Python function
- Output strictly following example patterns
- Correct handling of names

- Sample inputs and outputs

Prompt:
Write a python program for accepting a full name as input and formats it as "Last, First".Example formats: • "John Smith" → "Smith, John" • "Anita Rao" → "Rao, Anita"

Code:

```
#task3
full_name = input("Enter full name: ")

parts = full_name.split()

first_name = parts[0]
last_name = parts[-1]

formatted_name = f"{last_name}, {first_name}"
print(formatted_name)
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\sanjay\AI Assistant coding> & C:\Users\
oding/lab assignment4.3.py"
Enter full name: Sanjay kumar
kumar, Sanjay
PS C:\Users\sanjay\AI Assistant coding>
```

Approach:
In this python program we can learn how one short prompt works by giving multiple examples.

Task 4: Comparative Analysis – Zero-Shot vs Few-Shot

Scenario
Different prompt strategies may produce different code quality.

Task Description

- Use zero-shot prompting to generate a function that counts vowels in a string
- Use few-shot prompting for the same problem
- Compare both outputs based on:
  - Accuracy
  - Readability
  - Logical clarity

Expected Output

- Two vowel-counting functions
- Comparison table or short reflection paragraph
- Conclusion on prompt effectiveness

Prompt:
Write a python program that counts vowels in a string and compare both outputs based on Accuracy,Redability and logical clarity.

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Code:

```
#task4
text = input("Enter a string: ")

vowels = "aeiouAEIOU"
count = 0

for ch in text:
    if ch in vowels:
        count += 1

print("Vowel count:", count)
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\sanjay\AI Assistant coding> & C:\Users\
oding/lab assignment4.3.py"
Enter a string: sanjay
Vowel count: 2
PS C:\Users\sanjay\AI Assistant coding>
```

Approach:
In this code we can know ,that how to conver vowels to string by using Zero short prompting means without giving any example.

Task 5: Few-Shot Prompting – File Handling
Scenario
File processing requires clear logical understanding.
Task Description
Use few-shot prompting to generate a Python function that:

- Reads a .txt file
- Counts the number of lines in the file
- Returns the line count

Expected Output

- Working Python file-processing function
- Correct line count
- Sample .txt input and output
- AI-assisted logic explanation

Prompt:
Write a python program by using few short prompting which Reads a .txt file ,counts the number of lines in the file and returns the line count.

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Code:

```python
#task5
def count_lines_in_file(file_path):
    with open(file_path, 'r') as file:
        return sum(1 for _ in file)
if __name__ == "__main__":
    sample_file = 'sample.txt'

    # Create a sample file
    with open(sample_file, 'w') as f:
        f.write("Hello, World!\n")
        f.write("This is a sample file.\n")
        f.write("It has multiple lines.\n")

    # Count lines
    line_count = count_lines_in_file(sample_file)
    print(f"The file '{sample_file}' has {line_count} lines.")
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    POR

PS C:\Users\sanjay\AI Assistant coding> & C:\Us
oding/lab assignment4.3.py"
The file 'sample.txt' has 4 lines.
PS C:\Users\sanjay\AI Assistant coding>
```

Approach:
In this code we can learn how the file is reading by using few short prompting.

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.